Lab 3
Motion Analysis using SolidWorks

Objective

The objective of this exercise is to learn about the Motion Analysis toolbox in SolidWorks. SolidWorks Motion for kinematic and dynamic analysis of mechanical systems is introduced. Two simple examples, i.e., a sliding block and a single-piston engine, are included that should provide a good overview of the motion analysis capabilities offered by SolidWorks Motion.

Example 1 – Sliding Block

The physical model of the sliding block is very simple. The block is made of cast alloy steel with a size of 1×1×1 in. As shown in **Error! Reference source not found.**, the block travels a total of 9 in. on the slope surface due to gravity. The units system employed for this example is IPS (in.-lbf-sec), in which the gravitational acceleration is 386 in./sec^2. The slider is released from a rest position at the top of the slope (that is, the initial velocity is 0). No friction is assumed between the block and the ground.

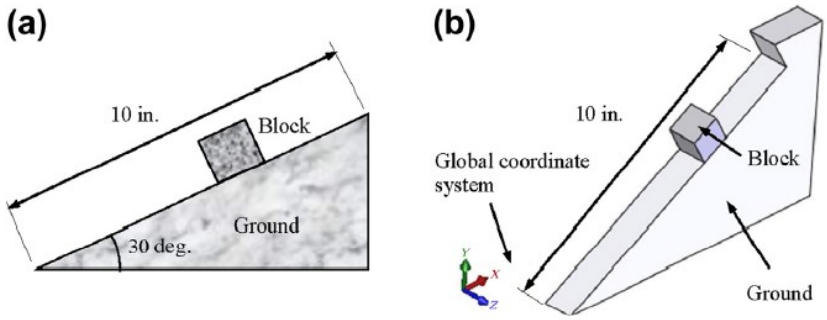


Figure 1: Sliding block. (a) Schematic view. (b) Motion model in CAD (SolidWorks).

For this lesson, the parts and assembly have been created in SolidWorks. There are four model files created: `block.SLDPRT`, `ground.SLDPRT`, `SlidingBlock.SLDASM`, and `Sliding-Blockwithresults.SLDASM`.

We start with `SlidingBlock.SLDASM`, in which the block is assembled to the ground and no motion entities have been added.

The assembly file `SlidingBlockwithresults.SLDASM` contains the complete simulation model with simulation results. In the assembly models, there are three assembly mates, as shown in **Error! Reference source not found.**:

- Coincident1(ground<1>, block<1>)
- Coincident3(ground<1>, block<1>)
- LimitDistance1(ground<1>, block<1>)

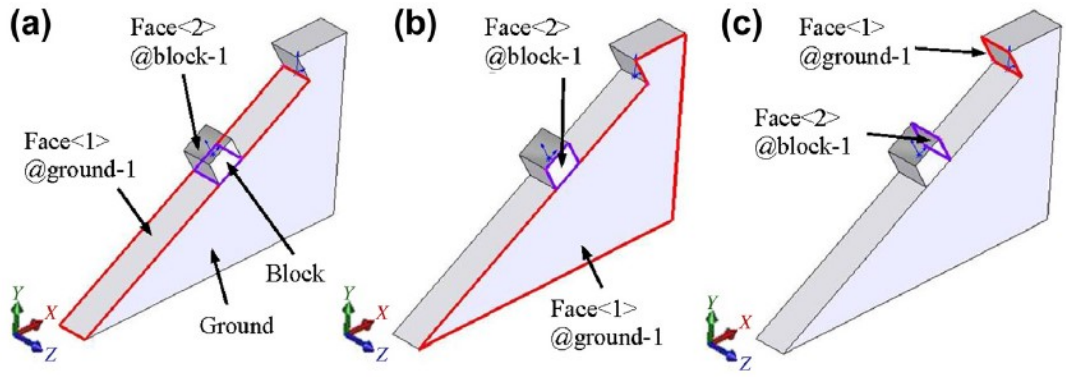


Figure 2: Assembly mates defined in `SlidingBlock.SLDASM`. (a) Coincident1. (b) Coincident3. (c) LimitDistance1.

The `LimitDistance` mate allows the block to move along the slope surface for a limited 9 in distance. By dragging the block in the graphics area, the block will move on the slope surface but not beyond it - because the `LimitDistance` mate does not allow this. Choose `Edit > Undo Move Component` from the pull-down menu to restore the block to its previous position. Now, consider the assembly mate `LimitDistance1`. From the SolidWorks browser, click `LimitDistance1`, and choose `Edit Feature`, as shown in Error! Reference source not found.. The mate is brought back for reviewing or editing.

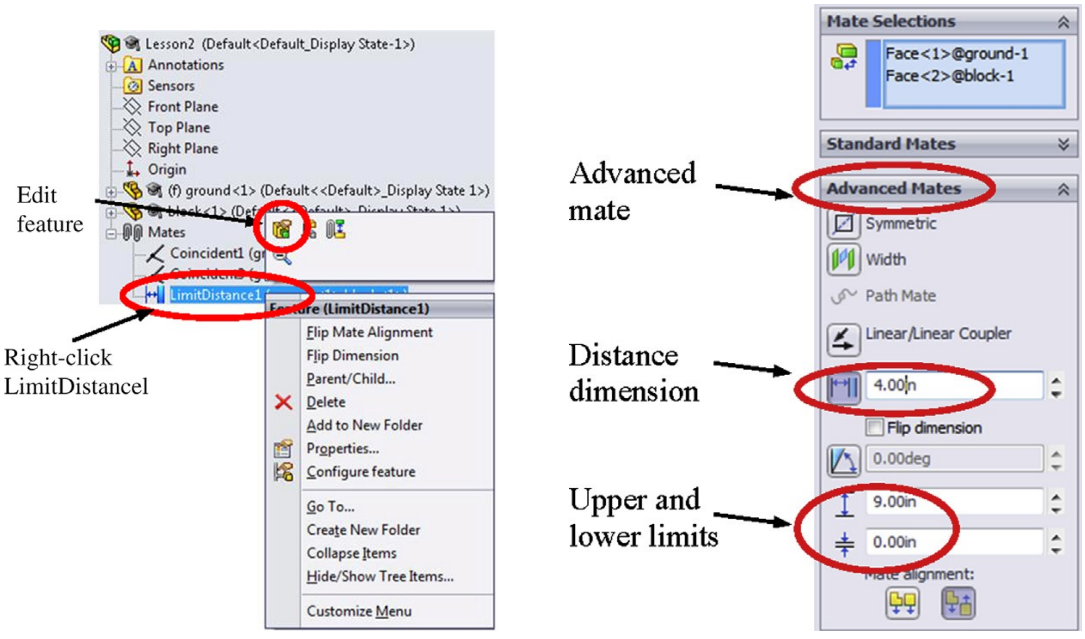


Figure 3: Mate Selection dialog box.

Note that the distance between the two faces, `Face<2>@block-1` and `Face<1>@ground-1` (**Error! Reference source not found.**c) is 4 in The upper and lower limits of the distance are 9 in. and 0 in, respectively. The length of the slope surface is 10 in; therefore, the upper limit is set to 9 in so that the block stops when its front lower edge reaches the end of the slope surface (since the block width is 1 in). Note that `LimitDistance1` is an advanced mate in SolidWorks.

Motion Model

The block and ground parts (or bodies) are assumed rigid. As mentioned earlier, a `LimitDistance` mate is defined to prevent the block from sliding out of the slope surface. The block reaches the end of the slope surface in about 0.3 sec, as shown in **Error! Reference source not found.**, which is the Y-position of the mass center of the block. The graph shows that it bounces back when it reaches the end; this action is due to the `LimitDistance` mate and is artificial. The Y-position of the mass center of the block is about 0.18 in., and travels down to about -4.31 in at 0.306 sec. A graph can be exported to an Excel file to check these numbers in SolidWorks Motion. The total vertical travel distance is 4.49 in.

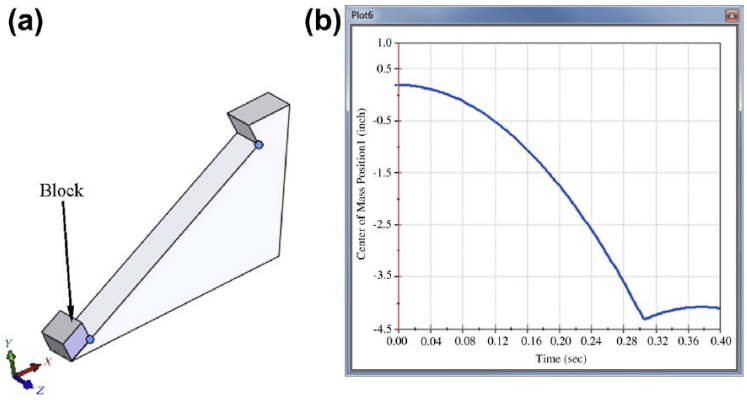


Figure 4: Sliding block in SolidWorks Motion. (a) Reaching the end of the slope surface. (b) Graph of the Y-position of the mass center of the block.

Defining Gravity

Click the `Gravity` button in the `Motion` toolbar to bring up the Gravity dialog box. Choose Y and keep the g-value (386.09 in./s2). In the graphics area, an arrow appears at the right lower corner, pointing downward to indicate the direction of the gravity. Click the checkmark on top of the dialog box to accept the gravity. A gravity node (Gravity) should appear in the `MotionManager` tree.

Defining Initial Position

Bring the block to the upper end of the slope surface as its initial position. The block is released from this position to simulate sliding without friction. Edite the mate `LimitDistance1` and enter 0 for

the distance dimension. From the SolidWorks browser, expand the Mates branch, click LimitDistance1(ground<1>,ball<1>), and choose Edit Feature . The definition of the assembly mate should appear in the dialog box (Error! Reference source not found.). Enter 0.00 in. for the distance and click the checkmark on top to accept the change. In the graphics area, the block should move to the top position on the slope face, as shown in Error! Reference source not found..

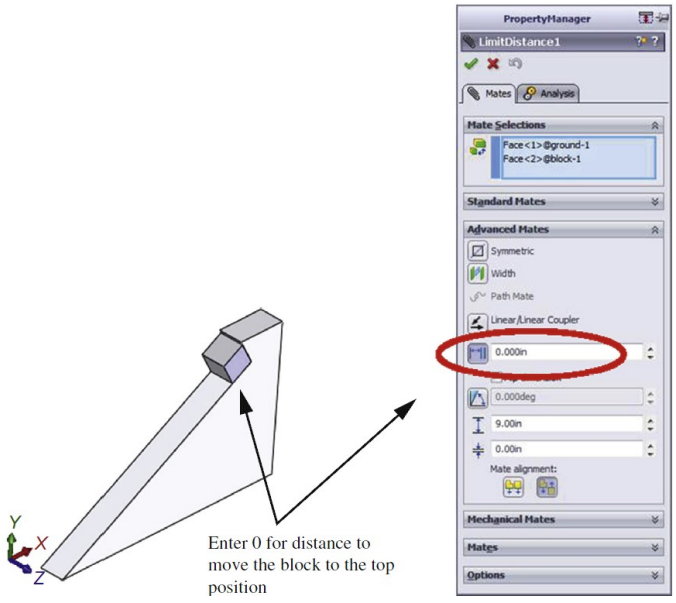


Figure 5: Define the initial position for the sliding block

Defining & Running the Simulation

Choose Motion Analysis from the motion study selection (directly above the MotionManager tree). Click the Motion Study Properties button from the Motion toolbar. In the Motion Study Properties dialog box, enter 500 for Frames per second, and click the checkmark at the top of the box. Zoom in to the timeline area until you can see tenth-sec marks. Drag the end time key to the 0.4-sec mark in the timeline area to define the simulation duration. Click the Calculate button on the Motion toolbar to start the motion analysis. A 0.4-sec simulation is carried out.

Displaying the Simulation Results

From the MotionManager, right click block<1>, and choose Create Motion Plot. In the Results dialog box, choose Displacement/Velocity/Acceleration, select Linear Displacement, Y Component, and then click the checkmark to accept the graph. You should see a graph like that in Figure S2.12b. Next, another graph is defined to show the same information.

Click the Results and Plots button from the Motion toolbar. In the Results dialog box (refer to **Error! Reference source not found.**), choose Displacement/Velocity/Acceleration, select Linear Displacement, and then Y Component. Pick the two vertices, as seen in **Error! Reference source not found.**. Click the checkmark to accept the graph. A graph like that of **Error! Reference source not found.** should appear. From the graph, the block moves along the slope surface from 0 (the distance between two coincided faces) to about -4 in. in about 0.3 sec. The block then "bounces" back because of the LimitDistance mate defined for the block. The graph does not seem to be correct since we know that the block will travel 4.5 in. in the Y-direction. This is because the distance the block travels is 9 in. and the slope is 30 deg. Therefore, the Y-distance the block travels should be 9 × sin 30 deg. = 4.5 in.

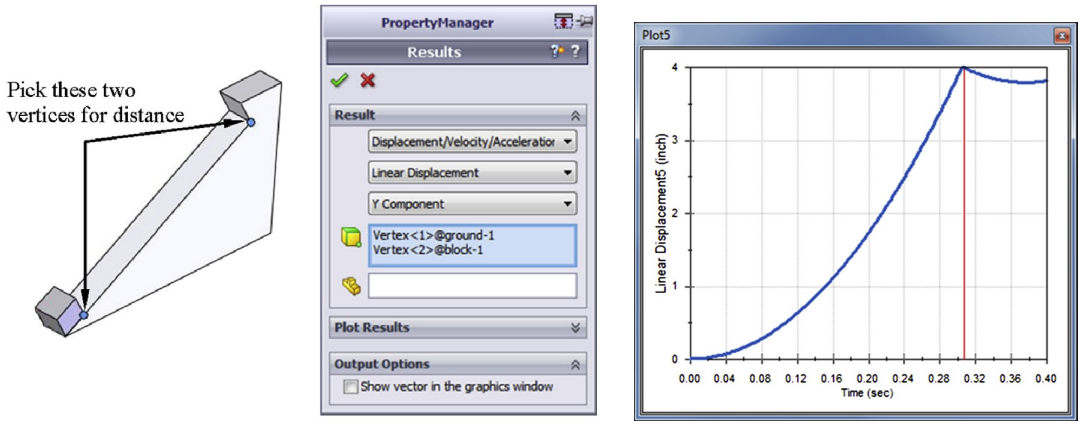


Figure 6: The Results dialog box - choosing vertices (left). Y-distance graph (right).

To examine the results in more detail, the graph data can be exported, for example, by right clicking the graph and choosing Export CSV. Open the spreadsheet and examine the data. As shown in the spreadsheet exported (**Error! Reference source not found.**), the time for the block to reach the lowest position is 0.306 sec, in which the Y-distance is about -4.5 in. This is accurate.

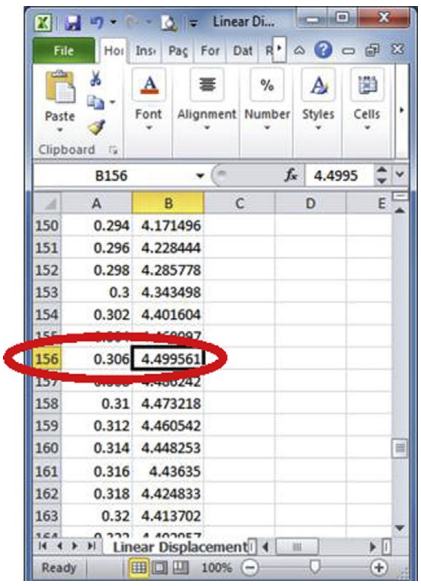


Figure 7: The calculation spreadsheet.

Lab write-up:
In your report, select a portion of your "design project" assembly where you can apply the "Motion Analysis". Pick an element and track its motion. Display the simulation results. Add discussion on how such results can help to better understand the dynamics of your system. Do you consider modifying your model?
Individual submission is required for this lab.